A Bilinear Isochronous Oscillator: Global Action–Angle Coordinates and Quantum Interface Spectrum

Source-local theorem reconstruction

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Abstract

We close the classical and quantum dynamics of a joined pair of quadratic half-wells. Every nonzero classical orbit has one energy-independent least period exactly when both stiffnesses are positive. Its global action–angle chart is C^1 and piecewise analytic, but detects unequal stiffness at second order. The natural Friedrichs quantization has simple discrete spectrum characterized completely by a parabolic-cylinder interface Wronskian. Flat-side, free, zero-energy, and equal-frequency boundaries are kept explicit.

1 The common-period owner

For $x_+ = \max\{x, 0\}$, $x_- = \min\{x, 0\}$, and $\omega_+, \omega_- \geq 0$, consider

$$H(x,p) = \frac{p^2}{2} + V(x), \qquad V(x) = \frac{\omega_+^2 x_+^2 + \omega_-^2 x_-^2}{2}. \tag{1}$$

The force is continuous and globally Lipschitz, hence the flow is unique and complete, and H is conserved.

Theorem 1 (all-energy isochrony)**.** *Every nonzero orbit of (1) is bounded and periodic with a common least period if and only if $\omega_+, \omega_- > 0$. In that chamber,*

$$T = \pi \left(\frac{1}{\omega_+} + \frac{1}{\omega_-} \right), \qquad J(E) = \frac{E}{2} \left(\frac{1}{\omega_+} + \frac{1}{\omega_-} \right), \qquad \Omega = \frac{dE}{dJ} = \frac{2}{1/\omega_+ + 1/\omega_-}. \tag{2}$$

Proof. At $E > 0$, write $B = \sqrt{2E}$ and start at $(0, B)$. The right half-excursion is

$$x = \frac{B}{\omega_+} \sin(\omega_+ t), \quad p = B \cos(\omega_+ t), \quad 0 \leq t \leq \frac{\pi}{\omega_+}.$$

Starting next from $(0, -B)$, with $s = t - \pi/\omega_+$, the left one is

$$x = -\frac{B}{\omega_-} \sin(\omega_- s), \quad p = -B \cos(\omega_- s), \quad 0 \leq s \leq \frac{\pi}{\omega_-}.$$

They give the least period in (2). Their two half-ellipse areas are $\pi E/\omega_+$ and $\pi E/\omega_-$, proving the action formula. If a frequency vanishes, an orbit entering that flat half-axis with nonzero speed escapes linearly; if both vanish the motion is free. This proves the necessity. $\square$